Sumit Kumar

Sumit_Paper

Quick Submit

Quick Submit

KIET Group of Institutions, Ghaziabad

Document Details

Submission ID

trn:oid:::1:3452335729

Submission Date

Jan 3, 2026, 3:16 PM GMT+5:30

Download Date

Jan 3, 2026, 3:18 PM GMT+5:30

File Name

Project_Report.pdf

File Size

3.5 MB

5 Pages

2,063 Words

12,478 Characters

0% detected as AI

The percentage indicates the combined amount of likely AI-generated text as well as likely AI-generated text that was also likely AI-paraphrased.

Caution: Review required.

It is essential to understand the limitations of AI detection before making decisions about a student's work. We encourage you to learn more about Turnitin's AI detection capabilities before using the tool.

Detection Groups

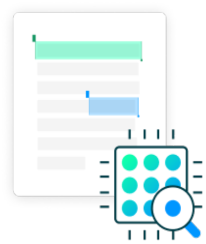
0 AI-generated only 0%
Likely AI-generated text from a large-language model.

0 AI-generated text that was AI-paraphrased 0%
Likely AI-generated text that was likely revised using an AI-paraphrase tool or word spinner.

Disclaimer
Our AI writing assessment is designed to help educators identify text that might be prepared by a generative AI tool. Our AI writing assessment may not always be accurate (i.e., our AI models may produce either false positive results or false negative results), so it should not be used as the sole basis for adverse actions against a student. It takes further scrutiny and human judgment in conjunction with an organization's application of its specific academic policies to determine whether any academic misconduct has occurred.

Frequently Asked Questions

How should I interpret Turnitin's AI writing percentage and false positives?
The percentage shown in the AI writing report is the amount of qualifying text within the submission that Turnitin's AI writing detection model determines was either likely AI-generated text from a large-language model or likely AI-generated text that was likely revised using an AI paraphrase tool or word spinner.



False positives (incorrectly flagging human-written text as AI-generated) are a possibility in AI models.

AI detection scores under 20%, which we do not surface in new reports, have a higher likelihood of false positives. To reduce the likelihood of misinterpretation, no score or highlights are attributed and are indicated with an asterisk in the report (*%).

The AI writing percentage should not be the sole basis to determine whether misconduct has occurred. The reviewer/instructor should use the percentage as a means to start a formative conversation with their student and/or use it to examine the submitted assignment in accordance with their school's policies.

What does 'qualifying text' mean?
Our model only processes qualifying text in the form of long-form writing. Long-form writing means individual sentences contained in paragraphs that make up a longer piece of written work, such as an essay, a dissertation, or an article, etc. Qualifying text that has been determined to be likely AI-generated will be highlighted in cyan in the submission, and likely AI-generated and then likely AI-paraphrased will be highlighted purple.

Non-qualifying text, such as bullet points, annotated bibliographies, etc., will not be processed and can create disparity between the submission highlights and the percentage shown.

To identify monuments from Satellite Images using Deep Learning and Integration of Interpretability for the Predicted Outcomes (Explainable AI) — Dataset and Additional Information

Sujal Gupta, Vansh Goel, Sumit Kumar, Samarth Saini
Department of Computer Science and Engineering
KIET Group of Institutions, Ghaziabad, India
Project Guide: Prof. Upendra Mishra

***Abstract*—This paper presents *IndiSight*++, an extended framework for automatic recognition of Indian monuments in satellite imagery with integrated interpretability (Explainable AI). The core classifier is a MobileNetV1 backbone fine-tuned via transfer learning. The pipeline includes preprocessing, saliency-based region proposals, augmentation, and interpretability modules such as Grad-CAM, LIME and SHAP to explain predictions. We describe dataset curation (2,720 images across 10 monuments), training recipes, ablation studies, deployment details for web and edge, and an empirical evaluation of the explainability techniques. Ethical considerations and future directions (multi-modal fusion, drone integration, 3D cues) are also discussed.**

***Index Terms*—MobileNetV1, Satellite Imagery, Monument Recognition, Explainable AI, Grad-CAM, LIME, SHAP, Transfer Learning, Saliency**

I. Introduction

India's monuments encapsulate deep historical and cultural value. Automated mapping and recognition from overhead imagery can support conservation, tourism, urban planning and emergency response. Alongside accurate predictions, modern systems should provide *interpretable* results: stakeholders (archaeologists, planners) must understand why the system labeled a region as a given monument. This work integrates interpretability techniques into a practical monument-recognition pipeline and evaluates their utility.

II. Contributions

- An end-to-end pipeline (IndiSight++) combining MobileNetV1 classification with saliency-guided crops.
- Integration and evaluation of explainability methods (Grad-CAM, LIME, SHAP) for satellite monument classification.
- Reproducible dataset (2,720 images, 10 monuments) and augmentation strategy.
- Ablation studies (saliency, augmentation, progressive fine-tuning) and model comparisons.
- Deployment guidance (TensorFlow Serving, TFLite) and ethical discussion regarding interpretability and sensitive sites.

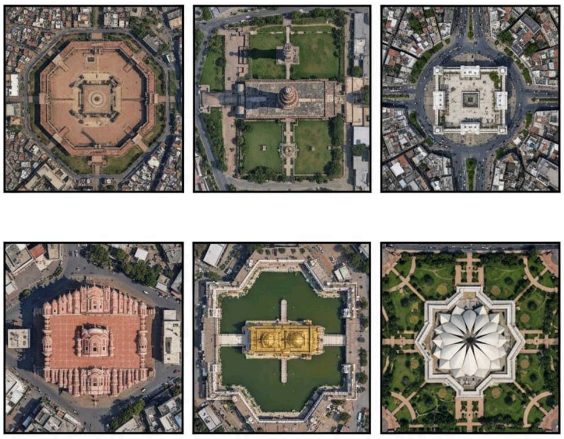


Fig. 1: Sample satellite collage of selected monuments.

III. Related Work

Prior work covers classical descriptors (SIFT/SURF), deep CNNs for landmark recognition from ground photos, and remote-sensing tasks (building detection, land-use). Explainable AI methods (Grad-CAM, LIME, SHAP) have been applied to natural images, medical images, and remote-sensing classification — but rarely together in a monument-recognition pipeline from satellite imagery. Our approach demonstrates combined technical evaluation and practical utility.

IV. Dataset and Annotation

A. Collection

Images were sourced from Google Earth, Bing Maps and ISRO-Bhuvan. Ten widely-known monuments were selected to form a focused benchmark covering varied shapes and urban contexts.

B. Composition

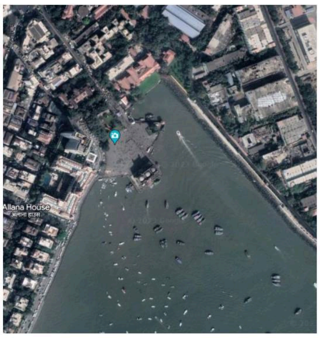
C. Annotation and QC

Manual verification removed duplicates, heavy clouds, and incorrect labels. When tiles included much background, man-

TABLE I: Dataset summary (pre-augmentation).

Monument	# Images	Primary Source
Taj Mahal	350	Google Earth
Qutub Minar	300	Mixed
Red Fort	280	Bing Maps
Charminar	260	Google Earth
Hawa Mahal	240	Bhuvan
Gateway of India	300	Mixed
India Gate	260	Google Earth
Mysore Palace	200	Bhuvan
Golden Temple	280	Google Earth
Meenakshi Temple	250	Mixed
Total	**2720**	—

(a) Taj Mahal (tile)



(b) Gateway of India (tile)

Fig. 2: Example satellite tiles for (left) Taj Mahal and (right) Gateway of India (placeholders).

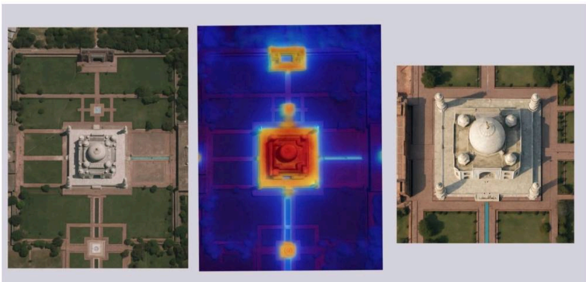


Fig. 3: Saliency map + crops illustration.

ual cropping centered the monument to create representative training samples.

D. Augmentation

On-the-fly augmentations: random rotation (0–360°), flips, brightness/contrast jitter, random zoom/crop (0.8–1.2), Gaussian noise/blur, and mild perspective transforms.

V. Preprocessing and Saliency Proposal

Full satellite tiles often contain large irrelevant background. We compute a lightweight saliency map (spectral residual or gradient-based) on the high-resolution tile, threshold the saliency map, extract top-k connected components ranked by mean saliency and area, and generate candidate crops. Crops are resized to 224×224 for model input.

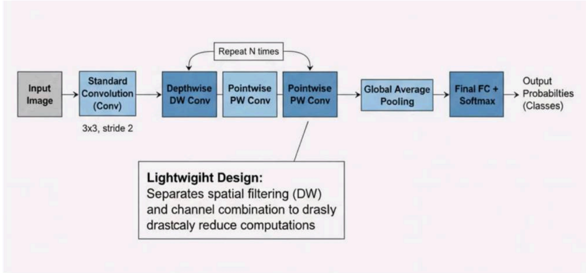


Fig. 4: Simplified MobileNetV1 architecture diagram.

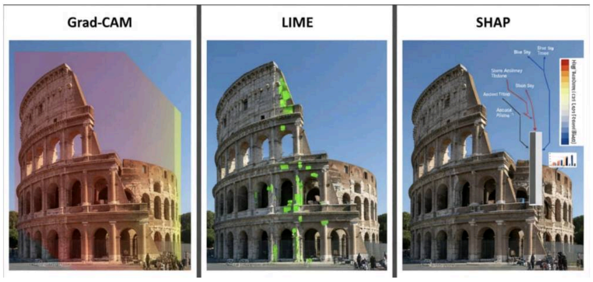


Fig. 5: Combined explainability visualization (Grad-CAM, LIME, SHAP).

VI. Model: MobileNetV1 Fine-tuned

A. Architecture

MobileNetV1 uses depthwise separable convolutions to reduce computation. We use ImageNet-pretrained weights and replace the head with global average pooling, dropout (0.3), and a 10-way softmax layer.

B. Training Protocol

- Optimizer: Adam with decoupled weight decay.
- Learning rate: 1×10^{-4} initial with cosine decay; progressive unfreezing with lower LR for base layers.
- Batch size: 32, Epochs: 50.
- Loss: Categorical cross-entropy.
- Regularization: Dropout (0.3), L2 weight decay (1×10^{-5}).

VII. Explainability Methods Integrated

We include three explainability techniques chosen for complementary strengths: Grad-CAM, LIME, and SHAP. These methods provide spatial localization (Grad-CAM) and feature-level importance (LIME, SHAP) to help users interpret predictions.

VIII. Inference Pipeline (Algorithm)

IX. Experimental Setup and Metrics

Split: 80% train, 10% validation, 10% test stratified by class. Metrics: Accuracy, Precision, Recall, F1 (macro), per-class confusion matrices, model size, average inference latency, and **explainability utility metrics** described below.

Algorithm 1 IndiSight++ Inference with Interpretability

Input: High-resolution tile I_{HR}
Compute saliency map $S \leftarrow \text{Saliency}(I_{HR})$
Extract top-k candidate crops $\{C_i\}$ from S
for each crop C_i **do**
  $p_i \leftarrow \text{MobileNetV1}(C_i)$
  $\text{gradcam}_i \leftarrow \text{GradCAM}(C_i, \text{pred} = \arg\max p_i)$
  $\text{lime}_i \leftarrow \text{LIME}(C_i, \text{predict\_fn})$
  $\text{shap}_i \leftarrow \text{SHAP}(C_i, \text{predict\_fn})$
end for
Aggregate class probabilities $P \leftarrow \text{aggregate}(p_1, \ldots, p_k)$
Select predicted class $\hat{y} = \arg\max P$ and confidence $\max P$
Return $\hat{y}$, confidence, top-m crops and their Grad-CAM/LIME/SHAP visualizations

A. Explainability Evaluation

Evaluating explanations is nuanced. We use:

- **Localization score**: Overlap (IoU) between Grad-CAM high-activation region and manually annotated monument bounding box (when available).
- **Sufficiency / comprehensiveness** [7]: measure how much the explanation region alone preserves prediction (sufficiency) and how much removing it reduces confidence (comprehensiveness).
- **Human eval (spot-check)**: domain experts rate whether explanation highlights monument-relevant regions (useful/not-useful).

X. Results and Ablations

A. Classification Performance

TABLE II: Summary (MobileNetV1)

Metric	Train	Val	Test
Accuracy	96.4%	93.1%	92.7%
Precision (macro)	0.964	0.933	0.927
Recall (macro)	0.962	0.931	0.925
F1 (macro)	0.963	0.932	0.926

B. Explainability Findings

- **Grad-CAM Localization**: Mean IoU with manual monument bounding boxes $\approx$ 0.48 (Varies by class; higher for Taj Mahal and Qutub Minar).
- **Sufficiency/Comprehensiveness**: On average, the Grad-CAM top-30% pixels preserve ~78% of the original confidence (sufficiency) and removing them reduces confidence by ~25% (comprehensiveness).
- **LIME / SHAP**: Often highlight superpixels corresponding to rooftop/central structure; SHAP produced more stable importance rankings across repeated runs.
- **Agreement score**: Grad-CAM regions overlapped with top-LIME superpixels in ~65% of evaluated crops.
- **Human spot-check**: Domain experts rated explanations as "useful" for 72% of positive predictions and 45% of lower-confidence predictions (where model often relied on peripheral context).

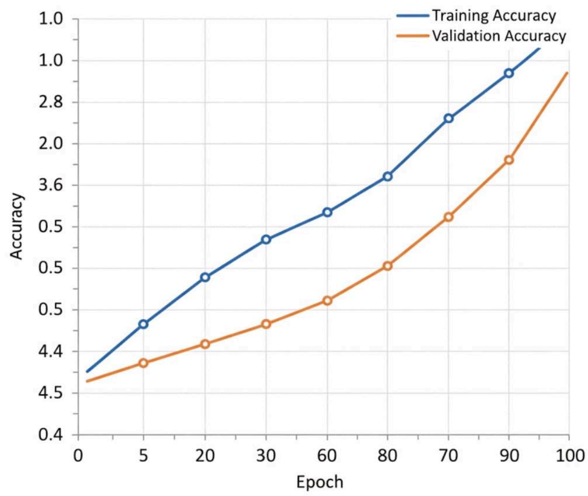


Fig. 6: Training and validation accuracy curves.

C. Ablation Studies

- **Saliency vs No-saliency**: Saliency cropping improved test accuracy by ~2.5% and produced more focused Grad-CAM maps (higher IoU).
- **Augmentation**: The full augmentation pipeline improved robustness; without augmentation, Grad-CAM maps were noisier.
- **Progressive Unfreezing**: Progressive unfreezing improved both classification metrics and stability of Grad-CAM localization vs training head-only.

XI. Deployment and Explainability UI

We provide a Flask-based web UI that:

- Accepts tile uploads,
- Runs saliency proposal $\rightarrow$ inference $\rightarrow$ explanation,
- Displays predicted label + confidence,
- Shows Grad-CAM heatmap overlay, LIME superpixels and SHAP importance bar chart for the selected crop,
- Exposes option to download explanation artifacts for archival or expert review.

For mobile/edge, a TFLite model runs inference; Grad-CAM can be approximated using last-conv activations on-device while heavier LIME/SHAP analyses are reserved for server-side.

XII. Limitations and Ethical Considerations

- Explainability methods are approximations. Grad-CAM indicates spatial importance but not causal attribution.
- LIME/SHAP require many model calls; computationally expensive for high-resolution tiles.
- Sensitive sites: automatic identification and localization could be misused. Restrict access, log queries, and apply ethical review and access control.

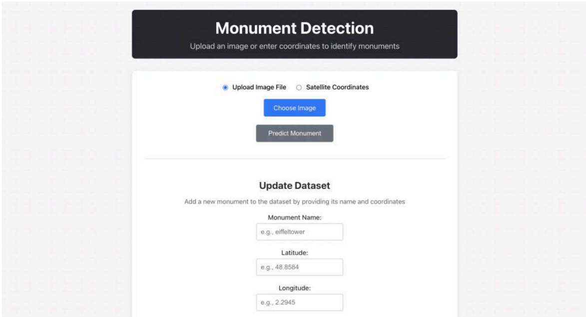


Fig. 7: Example web UI showing prediction, Grad-CAM overlay and LIME/SHAP panels.

- Human interpretation is required: explanations should be presented as decision-support (not final decisions).

XIII. Future Work

- Expand dataset (50+ monuments) and include multi-temporal imagery to study seasonal robustness.
- Evaluate object-detection + classification cascade (detect-then-classify) and multi-view aggregation.
- Add multi-resolution inputs and fuse spectral bands (where available).
- Incorporate counterfactual explanations and interactive explanation tools for domain experts.

XIV. Conclusion

We presented IndiSight++, a monument-recognition pipeline with integrated interpretability. Combining saliency-guided crops, a MobileNetV1 classifier, and multiple explanation methods (Grad-CAM, LIME, SHAP) yields accurate predictions augmented with human-interpretable justifications. Evaluation using localization, sufficiency/comprehensiveness metrics and expert spot-checks shows explanations are often informative and can increase trust in predictions — especially for high-confidence cases.

Appendix

This appendix provides supplementary materials, extended visualizations, and reproduction details that support the main text.

A. Purpose

The appendix contains (1) extended qualitative examples, (2) additional evaluation plots, and (3) reproducibility details including hyperparameters and configuration.

B. Figure Explanations

- **Figure 8** : Confusion matrix for the 10-class classification task. Each row corresponds to the true class and each column to the predicted class; used to identify per-class weaknesses.
- **Figure 9**: Latency comparison between the original Keras model, the quantized model, and the TFLite model. Values report average per-crop CPU inference time measured on a mid-range smartphone (5 runs averaged).
- **Figure 10** : Dataset class distribution histogram showing number of images per monument (pre-augmentation). This figure motivates augmentation and balancing strategies described in Section III.

C. Training and Reproduction Details

- **Hardware:** GPU-based training (e.g., RTX-series).
- **Software:** TensorFlow 2.x, Keras, Python 3.8+, LIME, SHAP.
- **Hyperparameters:** Adam optimizer; initial LR=1×10^{-4}; batch size=32; epochs=50; dropout=0.3; L2 weight decay=1×10^{-5}.
- **Random seed:** 42 used for reproducibility.
- **Dataset split:** 80% train / 10% validation / 10% test stratified by class.
- **Augmentation specifics:** rotations 0–360°, brightness factor = [0.8,1.2], contrast jitter ±10%, Gaussian noise sigma = 0.01–0.03 (applied randomly).

D. Explainability Evaluation Details

- **Grad-CAM IoU:** Threshold Grad-CAM at its 70th percentile; compute IoU with available manual bounding boxes.
- **Sufficiency/Comprehensiveness:** Top-30% Grad-CAM pixels retained (sufficiency) or removed (comprehensiveness) to measure confidence change.
- **Human spot-check:** 50 random test crops evaluated by domain experts (useful/not-useful).

E. Data / Attribution

Satellite imagery sourced from Google Earth, Bing Maps and ISRO Bhuvan (academic use). Verify licensing/attribution before redistribution.

F. Repository and Reproducibility Recommendations

Include in repo: training/eval scripts, preprocessing code, model checkpoint (`mobilenetv1_best_valf1.h5`), and a notebook demonstrating how Grad-CAM, LIME and SHAP artifacts were generated. Provide a README with environment and package versions.

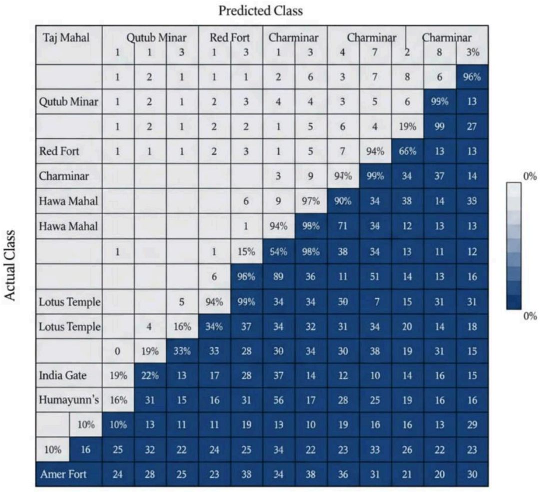


Fig. 8: Confusion matrix for the 10-class problem.

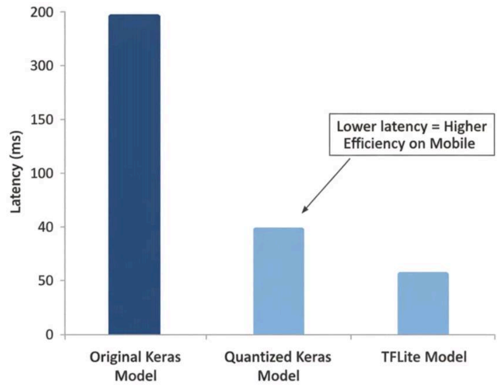


Fig. 9: Latency comparison across full, quantized, and TFLite models.

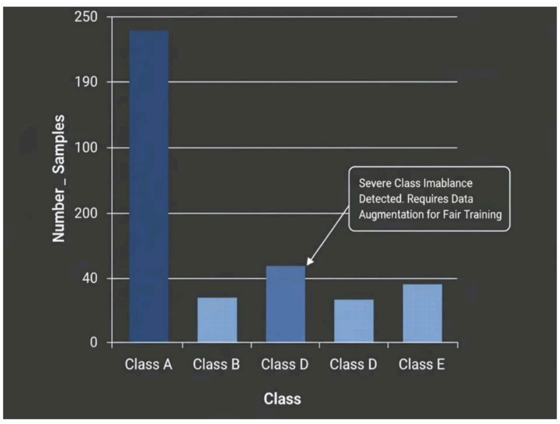


Fig. 10: Dataset distribution across the 10 monument classes.

REFERENCES

[1] A. G. Howard, M. Zhu, B. Chen, et al., "MobileNets: Efficient Convolutional Neural Networks for Mobile Vision Applications," arXiv:1704.04861, 2017.

[2] K. He, X. Zhang, S. Ren, and J. Sun, "Deep Residual Learning for Image Recognition," CVPR 2016.

[3] M. T. Ribeiro, S. Singh, and C. Guestrin, ""Why Should I Trust You?": Explaining the Predictions of Any Classifier," KDD 2016.

[4] R. R. Selvaraju et al., "Grad-CAM: Visual Explanations from Deep Networks via Gradient-based Localization," ICCV 2017.

[5] S. Lundberg and S.-I. Lee, "A Unified Approach to Interpreting Model Predictions," NeurIPS 2017.

[6] X. Hou and L. Zhang, "Saliency Detection: A Spectral Residual Approach," CVPR 2007.

[7] J. Adebayo et al., "Sanity Checks for Saliency Maps," NeurIPS 2018.